SUBMISSION PART 1 OF 3 · GROUP 16

Data Exploration & Preprocessing

Audit, cleaning, EDA and leakage-safe preparation of a 40,983-record PM2.5 station panel

This part documents where the data come from, what was removed and why, how the station-day panel behaves over time and space, and exactly how the modelling inputs are constructed so that no future information leaks into a forecast.

40,983	95	2,299	9
clean station-day records	coordinate-defined stations	distinct UTC daily dates	candidate environmental variables

Integrity note for the reader and examiner

Redistribution rights for the compiled Kaggle bundle are **unresolved**, and the publisher's full upstream row-inclusion rule cannot be reconstructed. Meteorology, trace gases and AOD are retained in the audit but **excluded from the model inputs** because their availability at the forecast origin is not documented. Both points are treated as first-class limitations, not footnotes.

NAVIGATION

Contents

Contents . . . 2
Objective and scope . . . 3
- What Part 1 must deliver . . . 3
- Research questions anchored in this part . . . 3

Dataset, licence and coverage . . . 4
- Dimensions and coverage . . . 4
- Variable inventory and missingness . . . 4
- Licence and provenance status . . . 4

Cleaning and quality controls . . . 5
- The cleaning rule, in one sentence . . . 5
- Checks performed and their outcomes . . . 5
- Station identity: why the count moves from 98 to 95 . . . 5

Target distribution . . . 6
- What the distribution implies . . . 6

Seasonality and time trends . . . 7
- Coverage is sparse and irregular — the single most important data fact . . . 7

Correlations and missing values . . . 8
- Association with the target . . . 8
- Missing-value profile . . . 8

Missing-value strategy . . . 9
- Rules . . . 9

Feature construction . . . 10
- The sequence contract . . . 10
- Feature channels . . . 10
- Three hard rules that prevent leakage . . . 10

Preprocessing pipeline . . . 11
- Leakage-control checklist . . . 11

Limitations and what Part 2 receives . . . 12
- Material limitations carried forward . . . 12
- The frozen handoff to Part 2 . . . 12

How Part 1 connects to the rest of the project

Part 1 fixes the **data contract**: the cleaned panel, the station-identity rule, the target bounds, the seven-day past-only sequence definition, and the fold-local preprocessing rule. Part 2 consumes exactly this contract to train, validate and test the CNN–LSTM; Part 3 consumes the resulting out-of-fold predictions for statistical testing and explainability. Every count and statistic used downstream originates in the audit files cited here.

Objective and scope

The project forecasts the next-day PM2.5 concentration at a monitoring station from observations available at the end of the current day. PM2.5 is fine particulate matter; the forecast target is a **continuous concentration in µg/m³**, so the task is regression, not classification. Part 1 does not model anything: it establishes that the data are trustworthy enough to model and that the inputs are constructed without leakage.

What Part 1 must deliver

- A verified dataset description: source, licence status, variable units, dimensions and coverage.
- A documented cleaning policy: exactly which records were removed, on what rule, and what was deliberately **kept** despite being unusual.
- A quantitative EDA: target distribution, seasonality, monthly structure, correlations, missingness and time-series behaviour.
- A missing-value strategy fitted inside each training fold rather than on the whole dataset.
- A precise feature definition: seven past-only daily steps per station, with no gap crossing and no future information.
- A leakage-control statement that downstream parts can rely on.

Research questions anchored in this part

RQ	Question owned or informed by Part 1
RQ1	Is the cleaned panel complete and consistent enough for matched-fold comparison against simple baselines?
RQ3	Which **available** feature groups carry signal, and which are excluded because they are unavailable at forecast time?
RQ4	How do seasonality, sparse coverage, extreme events, missingness and data rights constrain trustworthy use?

Parts 2 and 3 answer RQ1–RQ4 with fold-level evidence; Part 1 defines what "available" and "clean" mean.

Framing decision

The project is deliberately scoped to **one learned model family only** — CNN–LSTM — because that is Group 16′s assigned architecture. Part 1 prepares a single, reproducible input contract for it.

Dataset, licence and coverage

The active input is the publisher-prepared **Air Quality – Mediterranean Basin (2016–2026)** bundle (version 1), obtained from Kaggle and stored locally as `data/raw/extracted/combined_gee_pm25.csv`. Retrieval metadata, source checksums and the live Kaggle metadata snapshot are retained in `data/manifest.yaml` and `data/provenance/`.

Dimensions and coverage

41,718	40,983	95	0
raw combined rows	rows after cleaning	stations (lat/lon unique)	duplicate station-day keys

Attribute	Value	Unit	Source of record
Source	Kaggle · Sarah Ameziane, v1	—	`data/manifest.yaml`
Temporal span	2016-11-25 → 2026-04-06	UTC day	audit + EDA
Spatial span	lat 36.13–43.53 · lon −5.66–9.14	degrees	EDA station summary
Target	PM2.5 concentration	µg/m³	`Exploratory_Data.csv`
Target bounds	retain 0 < PM2.5 < 500	µg/m³	teacher cleaning rule
Rows removed	735 sentinel / out-of-rule	rows	audit

Variable inventory and missingness

Variable	Family	Unit	Missing	Role in the project
pm25	Target	µg/m³	0.00%	Forecast target
temperature_celsius	Meteorological	°C	0.00%	Excluded — timing unverified
pressure_mb	Meteorological	hPa	0.00%	Excluded — timing unverified
wind_u / wind_v	Meteorological	m/s	0.00%	Excluded — timing unverified
no2	Trace gas (GEE)	mol/m²	5.87%	Excluded — timing unverified
co	Trace gas (GEE)	mol/m²	5.87%	Excluded — timing unverified
o3	Trace gas (GEE)	mol/m²	5.97%	Excluded — timing unverified
aod	Satellite (GEE)	dimensionless	0.00%	Excluded — timing unverified

Only the PM2.5 target and calendar terms enter the model. Environmental columns are audited for completeness but withheld because forecast-origin availability is undocumented.

Licence and provenance status

Unresolved: redistribution rights

Kaggle labels the licence **“Other (specified in description)”**. The description points to OpenAQ, Copernicus and NASA sources but provides no compiled-bundle licence and no field-to-upstream attribution map. The bundle is therefore **not** presented as cleared for redistribution, and the original ETL is not recoverable. Sources are cited so a supervisor can reproduce the retrieval.

Cleaning and quality controls

The cleaning rule, in one sentence

Only rows whose PM2.5 value violates the teacher-specified physical rule were removed: records with PM2.5 ≤ 0 or PM2.5 ≥ 500 µg/m³ (735 rows). **No row was deleted for being statistically unusual** — extreme but physically plausible concentrations are retained, because statistical extremeness alone cannot establish an instrument fault.

735	727 / 8	0	0
rows removed (0 < y < 500)	low / high sentinel removals	out-of-rule values post-clean	invalid coordinates

Checks performed and their outcomes

Check	Result	Interpretation
Duplicate station-day keys	0	Each (station, date) pair is unique — the panel is a clean keyed table.
Out-of-rule / non-finite targets	0	After cleaning, every retained target is inside the physical rule.
Coordinate validity	0 invalid	All 95 stations carry finite latitude and longitude.
Station identity reconciliation	98 → 95	Three test entries removed by target cleaning; the identity logic is preserved.
Station-relative Tukey screen	352 flagged	Q3 + 3×IQR flags on 71 stations; flags are **retained for review**, never auto-deleted.
Development high-value counts	90 / 36 / 11	Values > 50 / > 100 / > 250 µg/m³ kept in the development partition.

Development-scope audit covers 30,651 rows and 73 stations up to 2024-12-07. See `results/16/cleaning_quality_audit/` and `results/16/audit.json`.

Station identity: why the count moves from 98 to 95

A station ID is built from the source sensor name plus latitude and longitude rounded to six decimals. This keeps two separately located instruments apart even when a sensor name repeats. The audit reconciles 98 numeric source identifiers to 98 coordinate-defined stations; cleaning removes three test entries, leaving 95. There are 94 distinct sensor names, and four of them map to two coordinate-defined stations each.

> **Why this matters for validation**
>
> Because a **name** is not a **station**, station-grouped cross-validation in Part 2 must split on the composite station ID (name + coordinates). Splitting on the name alone would leak the same instrument across train and test and inflate spatial scores.

Target distribution

The cleaned PM2.5 target is strongly right-skewed: a long, low body of ordinary days and a thin tail of pollution episodes. This shape shapes every modelling decision that follows — why absolute error metrics dominate, why MAPE is unsafe, and why rare high-concentration days are hard to learn.

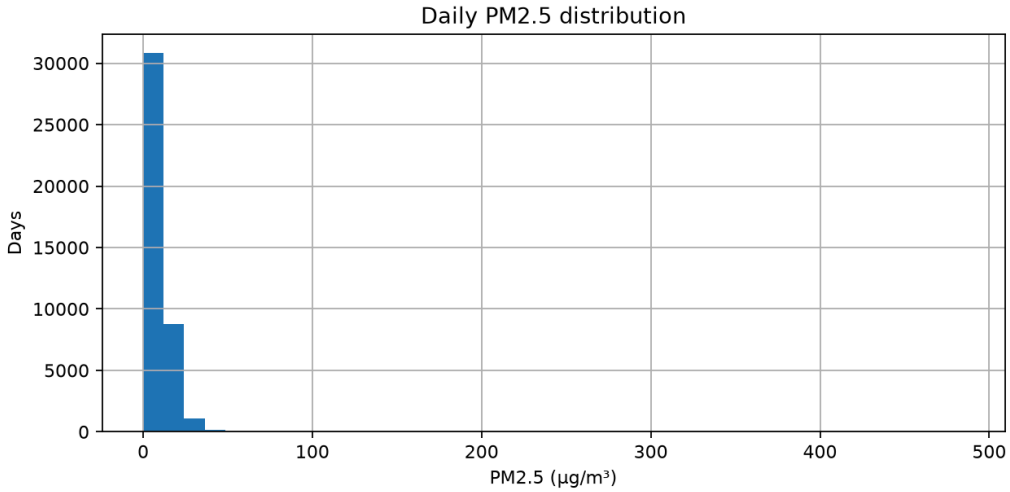


Figure 1. Distribution of cleaned daily PM2.5 (µg/m³) across 40,983 station-day records.

n	Mean	SD	Min	Median	P95	P99	Max
40,983	9.458	9.730	0.007	7.851	21.653	33.559	484.98

Summary statistics in µg/m³. Skewness 17.83 and excess kurtosis ≈ 649 confirm a heavy right tail; the maximum is ≈62× the median.

What the distribution implies

- Half of all station-days sit at or below **7.85 µg/m³**; the problem is mostly about ordinary days.
- The top 1% exceeds **33.6 µg/m³** and the maximum reaches **484.98 µg/m³** — roughly two orders of magnitude above the median.
- Because targets can approach zero, **MAPE is unstable**; RMSE, MAE and R^2 are the primary metrics in Parts 2 and 3.

Seasonality and time trends

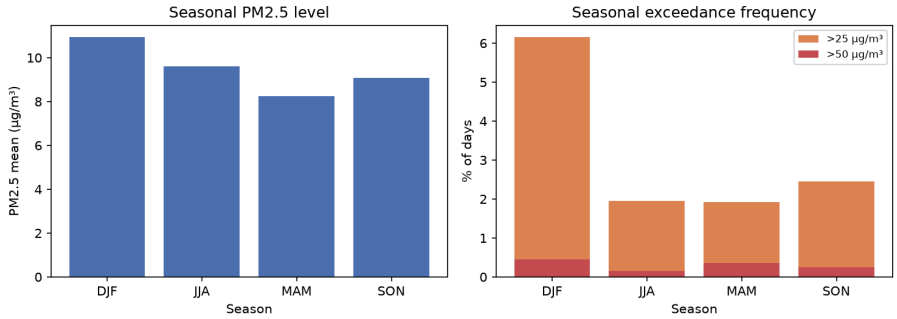


Figure 2. Seasonal PM2.5 summaries across the cleaned panel (DJF, MAM, JJA, SON).

Seasonal profiles

Season	Records	Mean	Med.	>25	>50
DJF	8,938	10.94	8.77	6.15%	0.45%
MAM	9,276	8.25	6.83	1.92%	0.36%
JJA	12,766	9.59	8.50	1.95%	0.15%
SON	10,003	9.08	7.47	2.46%	0.25%

Mean/median in µg/m³; percentages are record shares above the cutoffs — descriptive, not regulatory compliance rates.

Selected monthly means

Month	Mean	Med.	>25
Jan	10.39	8.15	4.85%
Feb	11.28	9.25	6.62%
Mar	9.49	7.26	4.00%
Jun	9.09	7.81	1.90%
Nov	10.17	7.91	3.59%
Dec	11.15	8.91	7.05%

Winter months (Nov–Feb) carry the highest means and the largest share of high days; full table in `eda/monthly_trend.csv`.

Coverage is sparse and irregular — the single most important data fact

The panel is **not** a dense grid. Dates and stations are present unevenly, which is why Part 1 measures behaviour on uninterrupted segments instead of pretending every calendar day is an observation.

Property	Value	Consequence
ACF/PACF availability	63 stations	Stations with ≥ 14 consecutive daily development observations.
Formal ADF / KPSS / Ljung-Box	6 stations	Only stations with ≥ 30 consecutive days can be tested; low-coverage diagnostics.
Lead–lag pollutant analysis	2–6 stations per lag	Descriptive only; no causal or availability claim.

Development partition only. Sequences in the model never bridge a date gap, so irregular coverage is handled by construction.

Correlations and missing values

Association with the target

All linear and rank associations between PM2.5 and the environmental variables are **weak**. The strongest is the rank correlation with NO_2 (Spearman ρ = 0.247); no pair reaches |r| > 0.9, so there is no multicollinearity red flag, and no environmental variable is a strong same-day predictor here.

Variable	Pearson r	Spearman ρ	|r| > 0.9?	Reading
no2	0.159	0.247	No	Strongest, but still weak in absolute terms
aod	0.094	0.180	No	Weak positive association
pressure_mb	0.113	0.167	No	Weak positive association
wind_u	−0.079	−0.164	No	Weak negative association
o3	−0.087	−0.135	No	Weak negative association
co	0.041	0.036	No	Essentially none
temperature_celsius	−0.006	0.064	No	Essentially none

*Computed on the cleaned panel (`results/16/correlations.csv`). Because availability at forecast time is undocumented, these associations are reported as **context**, not as model inputs.*

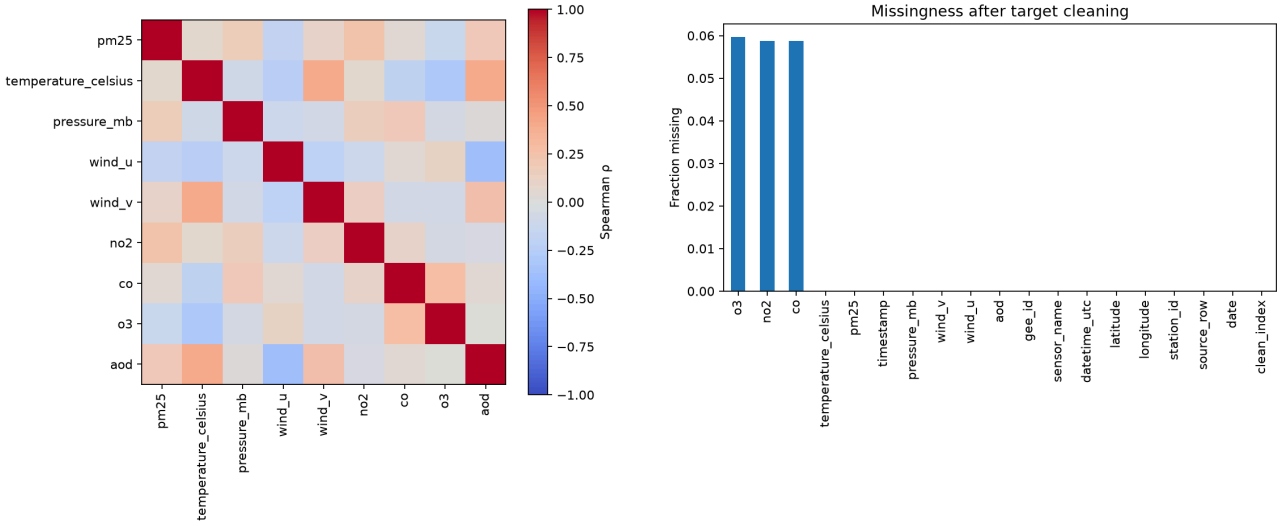


Figure 3 (left). Spearman rank-correlation structure. Figure 4 (right). Missingness profile of the cleaned table.

Missing-value profile

Column	Missing	Share	Handling decision
no2	2,407	5.87%	Not modelled; audited only
co	2,407	5.87%	Not modelled; audited only
o3	2,448	5.97%	Not modelled; audited only
pm25 and calendar	0	0.00%	Core modelling columns — no missingness

Missingness is not imputed globally

There is no dataset-wide fill. Any imputation happens **inside a training fold**, so the value used to fill a gap can never be computed from the validation or test period. This is a leakage control, not a convenience (Section 8).

Missing-value strategy

Because the **model** columns (PM2.5 target and calendar terms) are complete, the missing-value problem is small and confined to derived statistics. The strategy is therefore deliberately conservative.

Median	Fold-local	+ flags
imputation statistic	where it is fitted	indicator channels
robust to the heavy right tail	training partition only	record where a gap was filled

Rules

- **Median imputation** is used rather than the mean, because the target is heavily skewed and a mean fill would be dragged by episodes.
- The imputer and the scaler are **fitted on each training fold's sequences** and then applied to that fold's validation sequences — never fitted on the full panel.
- **Missingness indicators** are added for derived PM2.5 rolling summaries so the network can distinguish a real low value from a filled one.
- Standardisation (zero mean, unit variance) uses the same fold-local fit; the scalar output head is trained on the standardised target and predictions are inverse-transformed for scoring.
- Reading the target scale: all reported metrics are computed **after** inverse transformation, i.e. on the original µg/m³ scale.

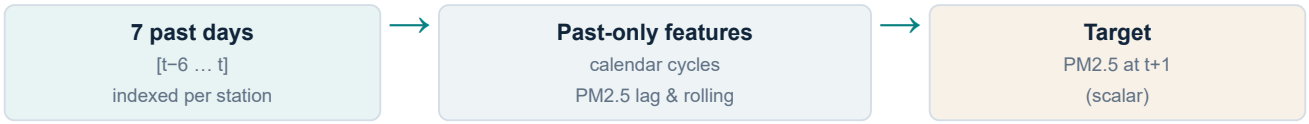
Why the reference preprocessing is frozen here

The target is kept on its original **µg/m³** scale, both rolling missingness indicators are retained, and imputation and scaling are fitted fold-locally. These choices are declared as the reference data contract so that every later stage consumes identical inputs. Their justification is leakage-safety and the strong right skew of the target — not model scores, which are outside Part 1′s scope.

Feature construction

The sequence contract

For station s and forecast origin day t, one training example is a **seven-day** window of daily rows for that station, ending at t, paired with the single target value at $t + 1$.



Feature channels

Channel	Type	Definition and why it is safe
pm25_lag1	Past value	PM2.5 observed at the forecast origin t — the persistence signal, and the strongest known predictor.
pm25_roll3_mean	Past rolling	Mean of the three days ending at t; uses only observed values ≤ t.
pm25_roll7_mean	Past rolling	Mean of the seven days ending at t; captures the recent level.
day_of_year_sin/cos	Calendar	Annual cycle encoded as a smooth pair; known in advance, so it cannot leak.
day_of_week_sin/cos	Calendar	Weekly cycle encoded as a smooth pair; known in advance.
rolling missing flags	Indicator	Declare where a rolling summary was filled, so filled and real values are distinguishable.

Nine processed channels in the reference model: seven features plus two rolling missingness flags. The channel set is fixed here as the reference contract consumed by all later stages.

Three hard rules that prevent leakage

- **No gap crossing.** A sequence is only built from consecutive daily rows; a missing date breaks the window instead of being skipped silently.
- **No station crossing.** A window never spans two stations, so one instrument's history cannot inform another's forecast.
- **No future information.** Every feature is a function of data at or before t; the label is a single value at $t + 1$. Automated tests mutate values at and after a cutoff and confirm earlier features do not change.

Preprocessing pipeline



Leakage-control checklist

Control	Status	Evidence
Fold-local imputation and scaling	Pass	Fitted on the training partition of each fold only.
Past-only rolling features	Pass	Recomputed from values ≤ the forecast origin.
Origin/target alignment	Pass	Each label is exactly the next station-day; verified by automated tests.
Gap and station boundaries	Pass	Sequence and split unit tests reject cross-station or gapped windows.
Environmental timing	Blocked	Exogenous channels withheld rather than assumed to be available at issue time.
Frozen holdout untouched	Pass	The reserved future block is excluded from all feature and model selection.

Verification boundary

The checks support leakage control in **the executed pipeline**. They do not prove that the publisher's original upstream join was leakage-free, nor reconstruct the complete row-inclusion process. That residual uncertainty is disclosed rather than hidden.

Limitations and what Part 2 receives

Material limitations carried forward

Area	Limitation carried into Parts 2 and 3
Licence	Compiled-bundle redistribution rights unresolved; no unrestricted-redistribution claim is made.
Upstream ETL	All 41,718 bundled rows map to source PM/GEE rows, but the publisher's omitted-row rule and tie handling remain unreconstructed.
Coverage	Sparse, irregular station-days limit formal temporal tests to six stations and make lead–lag analysis descriptive.
Environmental inputs	Meteorology, NO_2, CO, O_3 and AOD are withheld because forecast-origin availability is undocumented.
Tail behaviour	Only a handful of development observations exceed 100 µg/m³, so high-concentration performance is inherently unstable.

The frozen handoff to Part 2

40,983 × 9
cleaned panel
target + 8 audit variables

7 days
past-only sequence length

0 < y < 500
target rule (µg/m³)

Part 1 verdict

The panel is **clean enough to model and documented well enough to audit**. It is not a dense grid and its licence is unresolved, so Part 2 must (a) validate with expanding temporal folds and station-grouped folds, (b) keep preprocessing fold-local, and (c) report uncertainty honestly rather than a single number.

All Part 1 statistics are reproducible from `results/16/audit.json`, `results/16/Exploratory_Data.csv`, `results/16/Exploratory_temporal.csv`, `results/16/eda/`, `results/16/correlations.csv` *and* `results/16/cleaning_quality_audit/`.